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Improving Large Audio-Language Model Embedding Baselines for Instruction-Aware Speech Retrieval

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Abstract

Speech retrieval finds relevant spoken documents given a spoken query. The spoken query alone may not reveal the retrieval goal, such as finding an answer or matching speaker identity. Instruction-aware speech retrieval expresses this goal through a natural-language text instruction. Large audio-language models (LALMs) effectively process speech and text, and our analysis shows that different retrieval requirements favor hidden states from different audio-encoder layers. We introduce **IRIS-Embed**, which adapts LALMs into embedding models through contrastive training and instruction-aware audio layer weighting. We construct about 900K training samples from real data and synthetic data, with instructions expressing requirements for one or more speech attributes. On INSPIRE, an instruction-aware speech retrieval benchmark, IRIS-Embed consistently outperforms the corresponding LALM baselines. The best model achieves 62.92% macro-average Recall@50, surpassing the strongest baseline at 45.15%. Controlled comparisons demonstrate instruction awareness and the benefits of audio layer weighting. On MAEB, a massive audio embedding benchmark, macro-average gains of 9.4–19.8 points over the corresponding baselines show generalization beyond speech retrieval. The code and data are publicly available at https://github.com/lca0503/IRIS.

1 Introduction

Speech retrieval searches a collection of spoken documents using a spoken query (Chelba et al., 2008; Larson and Jones, 2012; Lee et al., 2015). For example, "Where should we go this weekend?" spoken excitedly could be used to find activity suggestions, another recording by the same speaker, or speech with similar excitement. The spoken query alone does not distinguish these goals.

Instruction-aware speech retrieval uses natural-language instructions to express the user's retrieval goal (Li and Lee, 2026). Each instruction expresses requirements for speech attributes such as spoken content, speaker identity, and speaking style. The required values may come from the spoken query or be specified directly. For example, "Find speech by the same speaker, spoken happily" asks for a document that matches the query's speaker identity and has a happy speaking style. A relevant spoken document must satisfy every requirement.

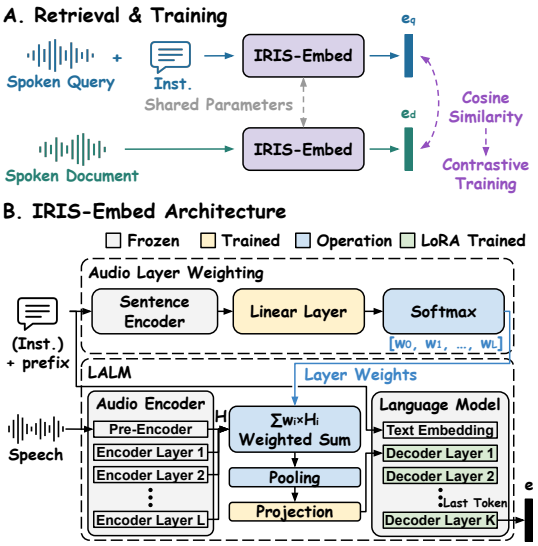

Figure 1: IRIS-Embed overview. A: Retrieval and training with a shared model. B: Audio layer weighting and embedding computation. Inst.: instruction.

Large audio-language models (LALMs) connect an audio encoder to a language model and have shown strong performance on tasks such as speech recognition, audio captioning, and audio question answering (Xu et al., 2025a; Team et al., 2026; Ghosh et al., 2026; Lu et al., 2026b). Their ability to process speech and text within a model motivates their adaptation for instruction-aware speech retrieval. Our analysis shows that different retrieval requirements favor hidden states from different audio-encoder layers. We ask: *How can we adapt LALMs for instruction-aware speech retrieval?*

We introduce IRIS-Embed (**I**nstruction-aware

Retr**I**eval for **S**peech), which adapts LALMs into embedding models through contrastive training and instruction-aware audio layer weighting. Figure 1 illustrates the overall framework. IRIS-Embed separately encodes a spoken query with its instruction and each spoken document, then ranks documents by cosine similarity. Audio-layer weighting predicts weights based on the instruction to combine hidden states from different layers of the frozen audio encoder before language-model encoding.

Contrastive training aligns query embeddings with relevant document embeddings. We construct about 900K samples from real data and synthetic data, each pairing a spoken query and instruction with positive and negative spoken documents. Each instruction expresses requirements for one or more speech attributes. The positive spoken document must satisfy every requirement in the instruction.

We evaluate IRIS-Embed on INSPIRE (Li and Lee, 2026), an instruction-aware speech retrieval benchmark. Across two model families, IRIS-Embed consistently outperforms the corresponding LALM baselines, achieving macro-average Recall@50 scores of 62.92% and 47.41%, both surpassing the strongest baseline at 45.15%. Controlled comparisons demonstrate instruction awareness for single- and multi-attribute speech requirements. We also show that audio-layer weighting improves speaker and speaking-style matching compared with using only the final audio-encoder hidden states. On MAEB (Assadi et al., 2026), we evaluate IRIS-Embed on 15 diverse tasks, including gender classification, accent-pair classification, and artist retrieval. IRIS-Embed achieves macro-average gains of 9.4–19.8 points over the corresponding LALM baselines, demonstrating generalization beyond speech retrieval.

2 Related Work

2.1 Spoken Content Retrieval

Spoken content retrieval finds spoken documents with content relevant to a spoken query (Garofolo et al., 2000; Chelba et al., 2008; Larson and Jones, 2012; Lee et al., 2015). Prior work explores query-by-example spoken term detection (Ao and Lee, 2018; Ram et al., 2018, 2020) and spoken passage retrieval for question answering (Lin et al., 2024a; Mundnich et al., 2025). We focus on instruction-aware speech retrieval, where instructions express requirements based on spoken content, speaker identity, speaking style, or other speech attributes.

2.2 Instruction-Aware Retrieval

Instructions guide text retrieval by expressing retrieval goals and relevance requirements (Asai et al., 2023; Su et al., 2023; Weller et al., 2025b,a; Wang et al., 2024; Lee et al., 2025a; Zhang et al., 2025b; Lee et al., 2025b). Image and multimodal retrieval methods also use text to express retrieval goals (Wu et al., 2021; Liu et al., 2021; Baldrati et al., 2023; Saito et al., 2023; Vaze et al., 2023; Zhang et al., 2024; Jiang et al., 2024; Zhang et al., 2025a; Zhou et al., 2025; Jiang et al., 2025; Meng et al., 2025; Li et al., 2026a). INSPIRE (Li and Lee, 2026) benchmarks instruction-aware speech retrieval with requirements for one or more speech attributes. We adapts LALMs into embedding models for instruction-aware speech retrieval.

2.3 Audio-Language Embeddings

Audio-language embeddings support retrieval between audio and text, as well as audio classification (Guzhov et al., 2022; Elizalde et al., 2023; Wu et al., 2023; Xu et al., 2023; Mei et al., 2024; Zhu et al., 2024; Li et al., 2026b). LALM embeddings also support retrieval between audio and text (Xiao et al., 2025; Chen et al., 2026; Yoo et al., 2026; Lu et al., 2026a). We focus on retrieving spoken documents given a spoken query and a natural-language text instruction. Furthermore, IRIS-Embed uses instruction-aware audio-layer weighting to combine hidden states from different layers of the frozen audio encoder before language-model encoding.

2.4 Audio Layer Aggregation

SUPERB aggregates frozen self-supervised speech-encoder hidden states for downstream tasks (wen Yang et al., 2021). Prompt-aware Mixture uses prompt-guided fusion for recognition and captioning (Shan et al., 2025). WAVE and OmniUE aggregate language-model states for multimodal embeddings (Tang et al., 2026; Wang et al., 2026). Instead, IRIS-Embed learns instruction-aware weights to combine hidden states from different layers of an LALM's frozen audio encoder through contrastive speech retrieval training.

3 Method

3.1 Problem Formulation

Instruction-aware speech retrieval ranks spoken documents in a collection D given a spoken query q and a natural-language text instruction z. Speech

attributes describe properties of a recording, such as spoken content, speaker identity, and speaking style. The instruction z expresses the requirements on these attributes by referring directly to the spoken query or specifying the values. For example, "Find speech by the same speaker, spoken happily" requires a document to match q's speaker identity and have a happy speaking style. A relevant document must satisfy every requirement; unconstrained attributes do not affect relevance.

For each query–instruction pair (q, z), let $D^+ \subseteq D$ denote the positive documents relevant to (q, z) and $D^- = D \setminus D^+$ denote the negative documents. A scoring function $f(d \mid q, z)$ ranks documents $d \in D$ in decreasing order of score. An ideal ranking satisfies $f(d^+ \mid q, z) > f(d^- \mid q, z)$, for every $d^+ \in D^+$ and $d^- \in D^-$.

3.2 Architecture

As shown in Figure 1, IRIS-Embed uses a shared model to separately encode the spoken query q with instruction z and each spoken document d. For each recording, we extract hidden states from the pre-encoder and all layers of the frozen audio encoder. The audio-layer weighting module encodes a text-only prompt to predict weights for combining these hidden states. The weighted audio hidden states are then pooled and projected before being passed to the language model, which processes the text and weighted audio hidden states using its own prompt template. Query prompts include instruction z for both audio layer weighting and language-model encoding, whereas document prompts contain no instruction z.

We use the language model's final-layer hidden state of the last non-padding token as the embedding, obtaining $\mathbf{e}_q$ for the query–instruction pair (q, z) and $\mathbf{e}_d$ for document d. Documents are ranked by decreasing cosine similarity, $f(d \mid q, z) = \cos(\mathbf{e}_q, \mathbf{e}_d)$. We next explain the motivation and design of audio layer weighting.

3.2.1 Audio Layer Weighting

To investigate how different audio-encoder layers perform across retrieval tasks, we evaluate their frozen hidden states on INSPIRE (Li and Lee, 2026). For each layer, we mean-pool the hidden states over time and rank spoken documents by cosine similarity to the spoken query, without using instructions, the audio projection, or the language model. Figure 2 reports Recall@50 for dialogue continuation, speaker matching, speaking-style matching, and answer retrieval. The best-performing layer varies across tasks, and the final layer does not consistently achieve the highest recall. These findings motivate combining hidden states from multiple audio-encoder layers before passing them to the language model.

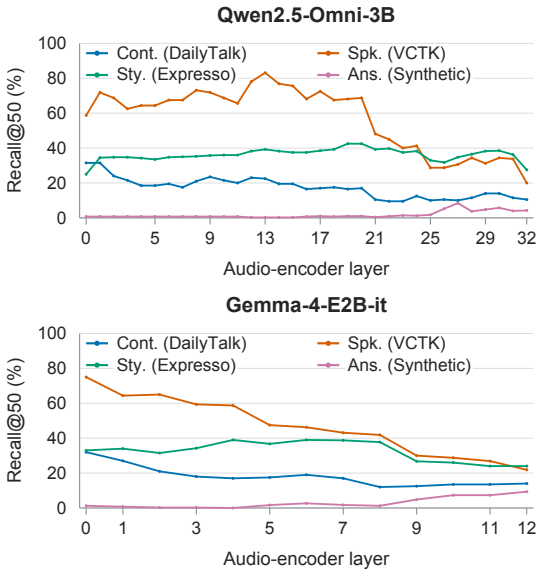

Figure 2: Recall@50 across frozen audio-encoder layers. Layer 0 represents pre-encoder hidden states; layers 1 and above represent encoder hidden states.

For each recording, let $\mathbf{H}_0$ denote the pre-encoder hidden states and $\mathbf{H}_1, \ldots, \mathbf{H}_L$ the hidden states produced by the L audio-encoder layers. We combine hidden states at the same time step to obtain weighted audio hidden states:

$$\widetilde{\mathbf{H}} = \sum_{\ell=0}^{L} w_\ell \mathbf{H}_\ell,$$

where w_ℓ is the scalar weight assigned to $\mathbf{H}_\ell$.

A shared audio layer weighting module computes weights from a text prompt p:

$$w_\ell = [\mathrm{softmax}(g(p))]_\ell,$$

where g uses a pretrained sentence encoder followed by a linear layer to produce $L + 1$ layer scores. The notation $[\cdot]_\ell$ selects the weight for $\mathbf{H}_\ell$. The resulting weights are nonnegative and sum to one. For queries, the prompt p includes the instruction z, allowing the layer weights to vary across retrieval tasks. For documents, we use a fixed prompt, so all documents share the same layer weights.

3.3 Training Objective

Each batch consists of B samples (q_i, z_i, d_i^+, d_i^-), where i indexes samples and d_i^+ and d_i^- are the paired positive and negative documents. The batch-

level document pool is $\mathcal{C} = \bigcup_{i=1}^{B} \{d_i^+, d_i^-\}$. Since documents from other samples may also be relevant to (q_i, z_i), treating them as negatives could introduce false negatives. We use training labels to mask these additional relevant documents. For each sample i, we construct a document pool $\mathcal{C}_i$ by excluding such documents from $\mathcal{C}$ while preserving the paired positive document d_i^+.

We optimize the model using the InfoNCE loss (Oord et al., 2018):

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp\big(f(d_i^+ \mid q_i, z_i)/\tau\big)}{\sum_{d \in \mathcal{C}_i} \exp\big(f(d \mid q_i, z_i)/\tau\big)},$$

where $\tau > 0$ is a temperature parameter controlling the sharpness of the score distribution. This objective encourages the model to assign higher relevance scores to the paired positive document than to the remaining negatives in $\mathcal{C}_i$.

During training, we freeze the audio encoder and the sentence encoder in g. We adapt the language model using low-rank adaptation (LoRA) (Hu et al., 2022). The audio projection, which maps audio hidden states to the language model's embedding dimension, is trained jointly with the added linear layer in g and the LoRA parameters.

4 Training Data

4.1 Notation

We represent each recording x by a set $A(x)$ of attribute-value pairs. Each pair identifies a speech attribute and its value, such as $(\text{style}, \text{happy})$ for a happy speaking style.

The requirements r form a set of attribute-value pairs that a relevant spoken document must satisfy. We write $r = r_q \cup r_s$. The requirements in r_q match the spoken query's attribute values, with $\emptyset \neq r_q \subseteq A(q)$. The set r_s contains attribute-value pairs with specified values that need not match the spoken query. We allow $r_s = \emptyset$. The text instruction z expresses r in natural language, referring to the spoken query for r_q and stating the values in r_s. The two sets use disjoint attribute keys to avoid conflicting requirements, so $r_q \cap r_s = \emptyset$.

For example, consider a spoken query q with happy speech and a dog barking in the background. The requirements $r = \{(\text{environment}, \text{dog bark}), (\text{style}, \text{sad})\}$ ask for sad speech with a barking dog. Here, $r_q = \{(\text{environment}, \text{dog bark})\}$ matches the query's environmental sound, while $r_s = \{(\text{style}, \text{sad})\}$ specifies a sad speaking style. A corresponding instruction z is "*Find a recording with the same background sound as the query, spoken sadly.*"

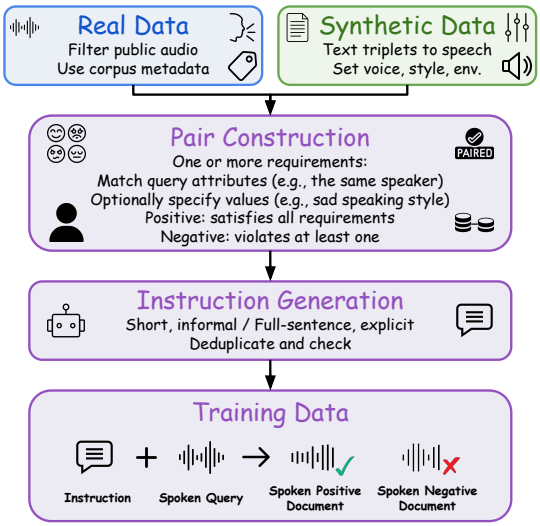

Figure 3: Overview of the data construction pipeline, combining real and synthetic data to create spoken query–document pairs and retrieval instructions.

4.2 Data Format

Each training sample (q, z, d^+, d^-) contains a spoken query, an instruction, and a positive and a negative spoken document. A document is relevant if it satisfies all requirements in r: $D^+ = \{d \in D : r \subseteq A(d)\}$. The paired positive belongs to D^+, while the paired negative belongs to D^- and violates at least one requirement. Attributes outside r do not affect relevance.

4.3 Construction

We construct IRIS-Embed training samples from real data and synthetic data, each paired with an instruction expressing its retrieval requirements. Figure 3 shows the data construction pipeline.

Real Data. We use public speech and audio corpora. The source annotations cover speaking style (Cao et al., 2014), speaker identity (Nagrani et al., 2017), gender (Zhou et al., 2021), accent (Zhao et al., 2018), sentiment (Poria et al., 2019), dialogue (Si et al., 2023; Park et al., 2024), and environmental sound (Salamon et al., 2014). We remove uninformative labels such as `unknown` and `other`, recordings outside 1–120 s, and environmental sounds containing speech. We also map equivalent labels across corpora, such as `happy` and `joyful`, to a common value. Corpus metadata provide attribute values for each recording, which we use to construct retrieval requirements and select query–document pairs.

Synthetic Data. We sample English text triplets from the MS MARCO (Nguyen et al., 2016) hard-triplet data, each containing a query, a relevant passage, and a hard negative passage. We synthesize speech from these texts using multiple text-to-speech models (Comanici et al., 2025; Hu et al., 2026). Voice selection controls speaker identity, synthesis prompts control speaking style, and sound mixing adds environmental sounds.

Pair Construction. For each spoken query q, we sample a nonempty r_q from $A(q)$ and optionally form r_s by sampling values for other attribute keys annotated for q. Speaker identity, dialogue, and answer relevance are restricted to r_q. When speaker identity is included, we exclude gender and accent, since matching the speaker also guarantees matching these attributes in our data. For each r, we sample a positive and a negative document, prioritizing negatives that violate fewer requirements to obtain harder negatives.

Instruction Generation. We use commercial large language models to express each set of requirements r as a natural-language instruction z. To cover diverse instruction styles, we use one prompt to generate short instructions and another to generate full-sentence instructions. We remove near-duplicates using ROUGE-L (Lin, 2004). We also verify that each instruction covers all requirements in r, refers to the spoken query for pairs in r_q, and specifies the values in r_s.

Statistics. The resulting dataset contains approximately 900K samples: 800K from real data and 100K from synthetic data. We use about 890K samples for training and 10K for validation. Appendix A lists the sources, counts, and data construction settings.

5 Experiments

5.1 Experimental Setup

Evaluation. We evaluate IRIS-Embed on four INSPIRE (Li and Lee, 2026) subsets. DailyTalk (Lee et al., 2023) evaluates dialogue continuation, VCTK (Yamagishi et al., 2019) evaluates speaker matching, and Expresso (Nguyen et al., 2023) evaluates speaker and speaking-style matching. Speaking style includes emotion and vocal states. The Synthetic subset evaluates answer retrieval and matching of speaker identity, speaking style, and environmental sounds, with instructions involving one or more attributes. Performance is measured by Recall@K for $K \in \{10, 50, 100\}$, defined as the fraction of relevant documents retrieved in the top K results, averaged across query–instruction pairs and reported as percentages.

Training. We train IRIS-Embed with Qwen2.5-Omni-3B (Xu et al., 2025a) and Gemma-4-E2B-it (Team et al., 2026) on the data in Section 4. The audio layer weighting module contains a sentence encoder and a learned linear layer. We use the frozen all-MiniLM-L6-v2 model (Wang et al., 2020; Reimers and Gurevych, 2019) as the sentence encoder. Training uses a global batch size of 512. Appendix B provides further training details.

5.2 Comparison Methods

Random. Uniform random document rankings serve as a chance-level baseline (Li and Lee, 2026).

General Speech Encoders. We evaluate general speech representations using the self-supervised WavLM-large (Chen et al., 2022) without instructions. Query and document embeddings are obtained by mean-pooling its final-layer hidden states over time.

Specialized Speech Encoders. We evaluate specialized speech representations without instructions, using ECAPA-TDNN (Desplanques et al., 2020) for speaker embeddings and emotion2vec-base (Ma et al., 2024) for emotion embeddings.

Cascaded Retrieval. We use a cascaded pipeline to retrieve spoken documents through text representations. Whisper-large-v3 (Radford et al., 2023) generates transcripts, while Qwen3-Omni-30B-A3B-Captioner (Xu et al., 2025b) generates captions. Qwen3-Embedding-8B (Zhang et al., 2025b) encodes the transcripts and captions of each spoken query and spoken document, with the instruction included in the query input.

LALM Embeddings. We evaluate Qwen2.5-Omni-3B (Xu et al., 2025a), Gemma-4-E2B-it (Team et al., 2026), and Qwen3-Omni-30B (Xu et al., 2025b) without additional embedding training. Embeddings are extracted from the final-layer hidden state of the last non-padding input token without generating a response. We also include LCO-Embedding-Omni-3B (Xiao et al., 2025), a Qwen2.5-Omni-based model trained for multimodal embeddings.

Method	Model	DailyTalk			VCTK			Expresso			Synthetic			Avg.		
		R@10	R@50	R@100	R@10	R@50	R@100	R@10	R@50	R@100	R@10	R@50	R@100	R@10	R@50	R@100
RAND	Random	0.24	1.06	2.10	0.35	1.84	3.63	0.30	1.50	2.96	0.20	1.00	1.98	0.27	1.35	2.67
ENC	WavLM-large	16.50	24.50	35.50	17.50	35.00	43.75	9.50	17.69	24.00	1.70	4.48	7.22	11.30	20.42	27.62
SENC	ECAPA-TDNN	15.50	24.00	29.50	**100.00**	**100.00**	**100.00**	**34.88**	41.75	45.44	10.02	14.83	18.82	40.10	45.15	48.44
	emotion2vec-base	11.50	23.00	30.50	3.13	6.25	13.75	0.88	4.38	8.00	1.61	4.53	6.82	4.28	9.54	14.77
CAS	Qwen3-Embedding-8B	**62.00**	**81.50**	**89.50**	1.88	6.25	10.63	0.75	5.56	9.69	7.02	8.95	11.07	17.91	25.57	30.22
LALM	Qwen2.5-Omni-3B	26.50	53.00	64.00	1.88	3.13	3.13	0.75	3.75	6.69	3.14	5.20	6.88	8.07	16.27	20.18
	Gemma-4-E2B-it	9.00	25.50	33.50	1.25	4.38	5.63	0.13	1.19	2.25	2.52	4.56	6.03	3.23	8.91	11.85
	Qwen3-Omni-30B	51.50	74.50	83.00	1.25	3.13	6.25	0.94	3.56	6.19	4.88	6.38	7.88	14.64	21.89	25.83
	LCO-Embedding-Omni-3B	54.00	75.50	83.50	1.25	2.50	5.63	1.94	8.44	13.81	6.82	9.37	11.65	16.00	23.95	28.65
IRIS	Qwen2.5-Omni-3B	60.00	77.50	87.50	67.50	85.00	89.38	29.63	**45.00**	**53.88**	**26.06**	**44.17**	**54.78**	**45.80**	**62.92**	**71.39**
	Gemma-4-E2B-it	25.50	44.50	54.50	58.75	74.38	79.38	28.63	43.81	52.31	15.07	26.94	36.08	31.99	47.41	55.57

Table 1: INSPIRE Recall@K. Avg.: macro-average across the four subsets. Bold / underline: best / second best. RAND: random; ENC / SENC: general / specialized encoders; CAS: cascaded retrieval; LALM: large audio–language models; IRIS: IRIS-Embed.

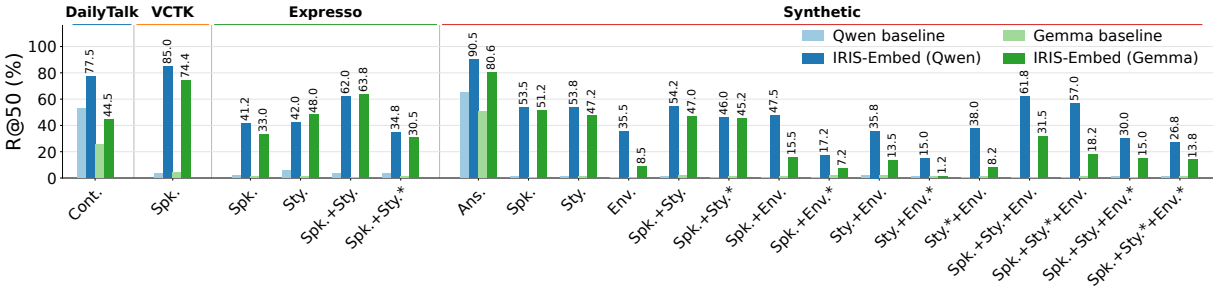

Figure 4: Recall@50 by instruction type on INSPIRE. Cont./Ans.: dialogue continuation/answer retrieval; Spk./Sty./Env.: speaker/speaking style/environmental sound. +: multiple attributes; ∗: specified values.

5.3 Main Results

Table 1 shows that IRIS-Embed consistently outperforms the corresponding LALM baselines on all four INSPIRE subsets. Macro-average Recall@50 reaches 62.92% for Qwen2.5-Omni-3B and 47.41% for Gemma-4-E2B-it, compared with 16.27% and 8.91% for their base models. Both exceed ECAPA-TDNN's 45.15%, the highest macro-average Recall@50 among comparison baselines.

Both IRIS-Embed models consistently outperform all LALM embedding baselines on VCTK and Expresso. IRIS-Embed with Qwen2.5-Omni-3B achieves 85.00% and 45.00% Recall@50 on these subsets, respectively, compared with 2.50% and 8.44% for LCO-Embedding-Omni-3B.

Both models also consistently outperform all comparison baselines on Synthetic. This subset includes instructions with multiple attributes, for which a relevant document must satisfy every requirement. Recall@50 reaches 44.17% for Qwen2.5-Omni-3B and 26.94% for Gemma-4-E2B-it, compared with 14.83% for ECAPA-TDNN.

On DailyTalk, IRIS-Embed with Qwen2.5-Omni-3B achieves 77.50% Recall@50, close to 81.50% for the cascaded pipeline using larger captioning and text-embedding models. While the specialized ECAPA-TDNN remains strongest for speaker matching on VCTK, IRIS-Embed supports retrieval across speech attributes within a single embedding model.

6 Analysis

6.1 Instruction Types

We breakdown IRIS-Embed's performance on instructions involving single or multiple attributes. Figure 4 compares IRIS-Embed with the corresponding LALM baselines across instruction types. Both IRIS-Embed models achieve higher Recall@50 on most instruction types.

On Synthetic, IRIS-Embed also performs well on instructions involving multiple attributes, including specified attribute values. For joint matching of speaker identity, speaking style, and environmental sound, Recall@50 reaches 61.8% for Qwen2.5-Omni-3B and 31.5% for Gemma-4-E2B-it. For the speaker matching with a specified speaking style, the scores reach 46.0% and 45.2%, respectively. Both models outperform all baselines on most multi-attribute instruction types. The complete results are provided in the Appendix D.

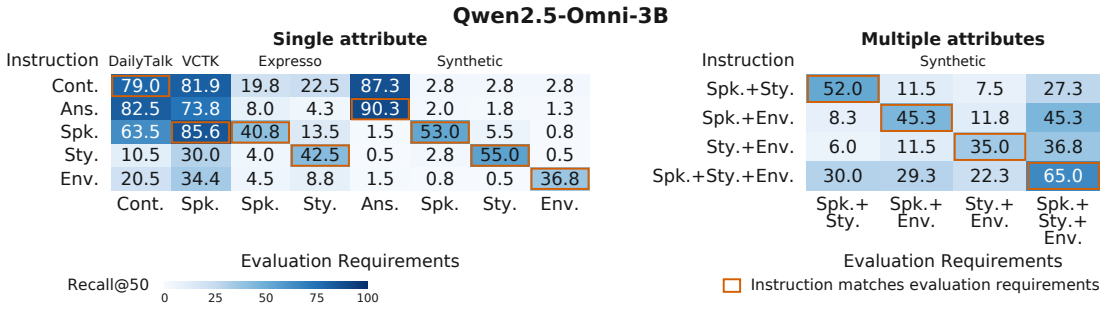

Figure 5: Recall@50 for IRIS-Embed with Qwen2.5-Omni-3B. Cont./Ans.: dialogue continuation/answer retrieval; Spk./Sty./Env.: speaker/speaking style/environmental sound; +: multiple attributes.

Method	DailyTalk	VCTK	Expresso	Synthetic	Avg.
Layer 13	15.50	76.88	**48.94**	35.27	44.15
Final Layer	**82.50**	48.75	28.25	34.72	48.56
Mean Layer	70.00	71.25	43.13	44.90	57.32
Finetuned	32.50	**91.25**	47.38	43.35	53.62
Static Weight	66.00	78.75	43.19	**45.28**	58.31
Inst. Weight (Ours)	77.50	85.00	45.00	44.17	**62.92**

Table 2: Recall@50 for Qwen2.5-Omni-3B audio layer aggregation on INSPIRE. Avg.: macro-average across the four subsets.

6.2 Instruction Awareness

We examine whether IRIS-Embed distinguishes different retrieval goals for the same spoken query. Figure 5 reports Recall@50 for Qwen2.5-Omni-3B under different short instructions, such as "Same speaker" and "Find the answer". Among five instructions with a single attribute, the instruction matching the evaluation requirements achieves the highest Recall@50 in seven of eight evaluation settings. Switching from a speaker-matching instruction to an answer-retrieval instruction increases answer retrieval Recall@50 from 1.5% to 90.3%.

On Synthetic, we compare four short instructions combining speaker, speaking style, and environmental sound, such as "Same speaker and same speaking style". The matching instruction achieves the highest Recall@50 in all four evaluation settings. Gemma-4-E2B-it shows a similar pattern, and both models also follow full-sentence instructions, as reported in Appendix D.2. These results show that IRIS-Embed follows the speech-attribute requirements expressed by the instruction.

6.3 Layer Aggregation

Table 2 compares Qwen2.5-Omni-3B variants trained and evaluated with different audio-layer aggregation methods. Layer 13 and Final Layer use the 13th and final audio-encoder hidden states, respectively. Mean Layer uses the average weighting across layers. Finetuned updates the audio encoder, while the other variants keep it frozen. Static Weight learns a single set of audio layer weights shared across all queries and documents, regardless of instructions or prompts. Inst. Weight is our instruction-aware audio layer weighting method, with weights varying according to the instruction.

Our method achieves the highest macro-average Recall@50, at 62.92%. Compared with Final Layer, our method improves recall on VCTK, Expresso, and Synthetic, increasing the macro-average by 14.36 points. Layer 13 and Finetuned improve speaker and speaking-style matching over Final Layer, but have much lower dialogue-continuation recall. These results suggest that retrieval tasks benefit from hidden states of different audio-encoder layers, highlighting the importance of instruction-aware audio layer weighting.

Although Mean Layer and Static Weight perform better on Synthetic, our method outperforms both on the other three subsets, achieving macro-average Recall@50 improvements of 5.60 and 4.61 points, respectively. These results demonstrate the effectiveness of instruction-aware weighting over instruction-independent layer aggregation.

With Gemma-4-E2B-it, our method also outperforms Final Layer on VCTK, Expresso, and Synthetic, showing improvements across multiple retrieval subsets. Appendix D.3 provides the full results for both models. Together, these results demonstrate the effectiveness of instruction-aware audio layer weighting in improving retrieval performance while keeping the audio encoder frozen.

6.4 Layer Weights

We analyze how audio layer weights vary across retrieval requirements. Figure 6 compares Qwen2.5-Omni-3B layer weights under five short instructions, such as "Same speaker" and "Find the answer", and the document prompt. To account for scale differences, we adjust the displayed weights

Method	Model	Clf										Clust		PC		Rrnk	Rtrvl	Avg.
		Beijing Opera	BirdCLEF CommonLang Age	GTZAN Genre	IEMOCAP Gender	MInDS14 Mridingham Tonic	SIBFLEURS VoxPopuli LanguageID					Vehicle Sound	VoxPopuli Gender	NMSQA VoxPopuli Accent		GTZAN	JamAlt Artist	
ENC	WavLM-large	57.7	19.8	16.9	67.7	63.1	13.5	31.4	12.5	89.0		0.2	0.1	49.0	52.2	68.8	78.5	41.4
CLAP	clap-htsat-fused	92.8	18.0	15.3	67.9	93.6	9.2	47.0	9.1	32.0		4.7	52.7	47.9	53.2	61.3	87.7	46.2
	larger_clap_general	93.6	17.0	**20.5**	84.5	89.3	8.0	49.4	9.4	84.6		3.4	3.3	47.4	53.5	66.8	**95.6**	48.4
LALM	Qwen2-Audio-7B	**97.4**	**37.1**	17.6	**93.1**	93.0	24.4	**61.2**	15.4	99.0		5.5	0.1	48.9	52.9	**80.8**	37.6	50.9
	LCO-Embedding-Omni-3B	93.2	31.6	16.9	81.0	59.4	**88.9**	39.5	**45.1**	63.4		2.2	1.0	**96.9**	50.7	75.4	67.1	**54.2**
	Qwen2.5-Omni-3B†	80.1	10.6	16.7	75.9	71.9	50.4	26.6	43.3	94.6		2.3	1.2	69.3	50.1	56.6	11.4	44.1
	Gemma-4-E2B-it†	63.6	4.0	15.5	28.2	50.5	22.0	27.0	21.9	**99.4**		1.2	0.1	57.2	50.2	38.2	17.8	33.1
IRIS	Qwen2.5-Omni-3B	91.9	13.3	14.3	80.1	98.7	24.4	33.5	20.9	71.0		1.1	55.2	92.9	61.5	66.1	77.6	53.5
	Gemma-4-E2B-it	88.5	15.1	16.4	60.6	**99.0**	14.2	32.3	26.6	79.2		**8.2**	**68.5**	94.3	**63.2**	60.6	67.0	52.9

Table 3: Scores on 15 MAEB audio tasks. Avg.: macro-average across the 15 tasks. Higher is better; bold marks column maxima. Clf: classification; Clust: clustering; PC: pair classification; Rrnk: reranking; Rtrvl: retrieval. CLAP: contrastive language–audio pretraining. †: baselines using the same task instructions as IRIS-Embed.

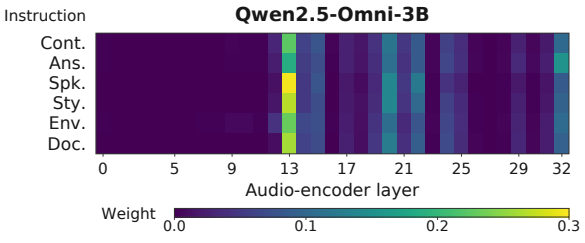

Figure 6: Scale-adjusted audio layer weights for Qwen2.5-Omni-3B. Doc.: document prompt; layer 0: pre-encoder hidden states; encoder layers start at 1.

using the root mean square of each layer's hidden states. This adjustment is used only for analysis. Layer 13 receives the highest adjusted weight across all five query instructions. Compared with speaker matching, answer retrieval places more weight on the final layer and less on layer 13. Gemma-4-E2B-it shows a similar pattern, with answer retrieval shifting weight from layer 4 toward its final layer. These weights show that instructions change the balance between intermediate and final audio-encoder hidden states. Appendix D.5 provides the full results for both models.

6.5 Task Generalization

We evaluate IRIS-Embed beyond retrieval tasks in INSPIRE on 15 MAEB (Assadi et al., 2026) audio tasks covering classification, clustering, pair classification, reranking, and retrieval. These include gender and accent tasks not covered by INSPIRE. We encode each recording with a task instruction using the query prompt, except for retrieval documents, which use the document prompt without an instruction. The models remain frozen during evaluation, and tasks derived from our retrieval training corpora are excluded. The corresponding LALM baselines receive the same task instructions. Appendix C.4 provides evaluation details.

Table 3 reports macro-average scores of 53.5 for Qwen2.5-Omni-3B IRIS-Embed and 52.9 for Gemma-4-E2B-it IRIS-Embed. These exceed the corresponding LALM baselines scores by 9.4 and 19.8 points, respectively, and approach LCO-Embedding-Omni-3B's 54.2. Both IRIS-Embed models outperform all comparison baselines on IEMOCAP gender classification, VoxPopuli gender clustering, and VoxPopuli accent pair classification. These results show that IRIS-Embed generalizes beyond instruction-aware speech retrieval.

7 Conclusion

IRIS-Embed adapts LALMs into embedding models for instruction-aware speech retrieval through contrastive training and instruction-aware audio layer weighting. Across two model families, macro-average Recall@50 on INSPIRE reaches 62.92% and 47.41%, compared with 45.15% for the strongest baseline. For Qwen2.5-Omni-3B, layer weighting improves VCTK Recall@50 from 48.75% to 85.00% over using only the final audio-encoder output. Controlled comparisons show that the models follow instructions with a single attribute or multiple attributes. On 15 MAEB tasks, macro-average scores improve by 9.4–19.8 points over the corresponding LALM baselines, showing generalization beyond speech retrieval. These results show that IRIS-Embed enables effective retrieval across speech attributes.

Limitations

Despite the advances achieved by our method, several limitations remain.

We prioritize smaller embedding models for computational efficiency and train IRIS-Embed with only two LALM backbones. Thus, its scalability to larger models remains unexplored.

IRIS-Embed combines hidden states from frozen pretrained audio encoders, making it dependent on the information they preserve. We do not explore alternative or complementary encoders that may capture broader speech attributes. Moreover, layer-weight visualizations do not establish individual layers' contributions to retrieval performance.

Our retrieval requirements rely on annotations covering limited speech attributes and values. Annotation and classifier errors may affect document-pair construction and in-batch masking, respectively. Our training also covers limited languages, voices, speaking styles, and environmental sounds, restricting its diversity relative to natural speech.

Finally, our cascaded baseline uses transcripts and audio captions. We leave comparisons with finer-grained cascaded approaches combining specialized speech embeddings or attribute-specific audio descriptions to future work.

While these limitations remain, IRIS-Embed demonstrates the effectiveness of adapting LALMs for instruction-aware speech retrieval. We believe our work will inspire further research on embeddings that follow diverse user intents and generalize across audio tasks.

Ethics Statement

We use public audio corpora and synthetic data, with corpus licenses listed in Table 4. Reuse should respect source licenses and speech-synthesis service terms, and shared synthetic recordings should be clearly labeled.

Speech-attribute matching can enable unauthorized speaker tracking or profiling. Applications should obtain appropriate consent, protect recordings and embeddings, and avoid discriminatory uses. Dataset biases may cause unequal retrieval quality, so deployment should include evaluation across speaker groups and recording conditions.

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A Data Construction

A.1 Data Sources

Table 4 summarizes the data sources, available annotations, and licenses for real and synthetic data. Speaking-style annotations include emotion and vocal states.

We use corpus metadata to construct retrieval requirements and select positive and negative spoken documents. Classifier predictions supplement these annotations only to identify and mask additional relevant in-batch documents during training, as described in Section 3.3.

A.2 Real Data

Preprocessing. We remove uninformative labels such as `unknown` and `other`, along with recordings outside 1–120 s. We exclude environmental-sound classes containing speech and map equivalent labels across corpora, such as `happy` and `joyful`, to a common value.

Classifiers. To supplement incomplete corpus annotations, we use audEERING's Wav2Vec2 age and gender model (Burkhardt et al., 2023)[1] to predict gender and emotion2vec+ large (Ma et al., 2024)[2] to predict emotion. For speakers with multiple recordings, we assign a shared predicted gender label through majority voting. Predicted labels are used only for relevance checks during training.

A.3 Synthetic Data

Text Selection. We sample 10,000 triplets from the MS MARCO (Nguyen et al., 2016) hard-triplet dataset processed by Sentence-Transformers.[3] Each triplet contains a query, a relevant passage, and a hard negative passage. To reduce false negatives, the source dataset uses the `ms-marco-MiniLM-L6-v2` cross-encoder[4] to score the top 50 BM25-retrieved passages per query. It retains negatives with scores more than 3.0 below the positive passage's score. We retain English queries and passages of 5–150 tokens and remove near-duplicates using ROUGE-L (Lin, 2004) with a threshold of 0.7.

[1] https://huggingface.co/audeering/wav2vec2-large-robust-24-ft-age-gender

[2] https://huggingface.co/emotion2vec/emotion2vec_plus_large

[3] https://huggingface.co/datasets/sentence-transformers/msmarco-bm25

[4] https://huggingface.co/cross-encoder/ms-marco-MiniLM-L6-v2

Speech Synthesis. We synthesize speech from these texts using Gemini 2.5 Pro TTS (Comanici et al., 2025)[5], Gemini 3.1 Flash TTS,[6] and Qwen3 TTS (Hu et al., 2026). Voice selection determines speaker identity. The two Gemini models share 30 voices, while Qwen3 TTS uses Ryan and Aiden, as detailed in Table 5.

Speaking Styles. Synthesis prompts control ten speaking styles: happy, sad, angry, fearful, surprised, disgusted, excited, sleepy, sarcastic, and whispering. Each prompt describes the voice, tone, pacing, and emotion. Table 6 provides the full prompts.

Environmental Sound. We use clips from eight UrbanSound8K (Salamon et al., 2014) classes: air conditioner, car horn, dog bark, drilling, engine idling, gunshot, jackhammer, and siren. Before cropping, we discard clips with root-mean-square levels below -45 dB relative to full scale. For each speech recording, we uniformly sample an environmental clip that passes this threshold.

Sound Mixing. We convert signals to mono by averaging channels and resample the environmental clip to match the speech sample rate. If the clip is longer than the speech, we randomly crop a segment matching the speech duration. Otherwise, we retain the full clip and place it at a uniformly sampled position within the speech. We uniformly sample a signal-to-noise ratio from $\{5, 10, 15\}$ dB and scale the environmental signal using the mean squared amplitudes of both signals over the mixing interval.

A.4 Pair Construction

We construct retrieval requirements and document pairs from the annotations described in Appendix A.1, following Section 4.2.

We balance sampling across attribute keys and values in the retrieval requirements. For each spoken query q, we sample a nonempty subset $r_q \subseteq A(q)$. To form r_s, we may select attribute keys annotated for q but not used in r_q. For each selected key, we sample a value to be specified in the instruction. Speaker identity, dialogue, and answer relevance appear only in r_q and refer to the

[5] https://blog.google/innovation-and-ai/models-and-research/google-deepmind/gemini-2-5-native-audio/

[6] https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-3-1-flash-tts/

Source	Attribute								License
	Answer relevance	Speaking style	Gender	Speaker	Accent	Sentiment	Dialogue	Environmental sound	
Real data									
VoxCeleb1 (Nagrani et al., 2017)		○	●	●					CC BY-SA 4.0
ESD (Zhou et al., 2021)		●	●	●					Custom
MELD (Poria et al., 2019)		●	○	●		●			GPL-3.0
CREMA-D (Cao et al., 2014)		●	●						ODbL 1.0
L2-ARCTIC (Zhao et al., 2018)		○	○	●	●				CC BY-NC 4.0
LibriSpeech-C (Lin et al., 2024b)		○	○					●	CC BY 4.0
EmoV-DB (Adigwe et al., 2018)		●	●	●					Custom
MSP-IMPROV (Busso et al., 2016)		●	●						Custom
VocalSound (Gong et al., 2022)								●	CC BY-SA 4.0
SpokenWOZ (Si et al., 2023)							●		CC BY-NC 4.0
MultiDialog (Park et al., 2024)							●		CC
UrbanSound8K (Salamon et al., 2014)								●	CC BY-NC 3.0
EMNS (Noriy et al., 2023)		●	○						Apache 2.0
Synthetic data									
MS MARCO (Nguyen et al., 2016)	●	●		●				●	CC BY 4.0

Table 4: Data sources and available annotations. ● indicates corpus metadata or synthesis annotations used to construct retrieval requirements and select positive and negative spoken documents. ○ indicates classifier predictions used only to mask additional relevant in-batch documents during training.

TTS model	Voice names
Gemini	Achernar, Achird, Algenib, Algieba, Alnilam, Aoede, Autonoe, Callirrhoe, Charon, Despina, Enceladus, Erinome, Fenrir, Gacrux, Iapetus, Kore, Laomedeia, Leda, Orus, Puck, Pulcherrima, Rasalgethi, Sadachbia, Sadaltager, Schedar, Sulafat, Umbriel, Vindemiatrix, Zephyr, Zubenelgenubi
Qwen3 TTS	Ryan, Aiden

Table 5: Synthesis voices used for synthetic data construction.

spoken query. When r includes speaker identity, we exclude gender and accent, since matching the speaker also guarantees matching these attributes in our data.

Pair Selection. For each spoken query and set of requirements r, we select a positive document satisfying all requirements and a negative document violating at least one. We prioritize documents that violate fewer requirements in r to obtain harder negatives.

A.5 Instruction Generation

We use Gemini 3.1 Pro Preview[7] and Claude Opus 4.6[8] to express each set of requirements r as natural-language instructions z. Two prompt variants produce short instructions and full-sentence instructions. We generate approximately 200 instructions per set of requirements before filtering and deduplication.

Prompt Variants. Table 7 presents the prompt templates, and Table 8 provides examples of generated instructions. Both templates distinguish query-matching requirements in r_q from specified values in r_s. The short-instruction template requests phrases of two to six words, while the full-sentence template requests conversational sentences. Both encourage diverse wording and avoid technical terms such as "criteria", "attribute", and "metadata".

Filtering. We retain instructions that express all requirements in r. Instructions must refer to the spoken query for r_q and specify the values in r_s. We allow synonyms and natural paraphrases of

[7] https://ai.google.dev/gemini-api/docs/models/gemini-3.1-pro-preview

[8] https://www.anthropic.com/news/claude-opus-4-6

Style	Prompt
Happy	**Voice:** bright, cheerful, and upbeat, as if you are smiling while you speak. **Tone:** warm, joyful, and openly affectionate. **Pacing:** lively and energetic, with buoyant rhythm. **Emotion:** genuinely delighted and optimistic, emphasizing key words with an uplifted intonation.
Sad	**Voice:** soft, low, and melancholic, with a gentle tremor. **Tone:** heavy, sorrowful, and subdued, as if holding back tears. **Pacing:** slow and drawn out, with thoughtful pauses. **Emotion:** deep sadness and quiet grief, letting the weight of the words settle.
Angry	**Voice:** sharp, intense, and forceful, with hard consonants. **Tone:** confrontational, tense, and biting. **Pacing:** clipped and driving, with emphatic stress on charged words. **Emotion:** strong anger held just under control, barely restrained frustration.
Fearful	**Voice:** trembling, breathy, and slightly higher in pitch. **Tone:** anxious, uncertain, and vulnerable. **Pacing:** uneven, with nervous hesitations and quick shallow breaths. **Emotion:** palpable fear, as if something dangerous is nearby and you are trying not to be heard.
Surprised	**Voice:** wide-eyed and astonished, with sudden lifts in pitch. **Tone:** amazed and caught off guard. **Pacing:** quick bursts punctuated by short breathy intakes. **Emotion:** genuine shock and disbelief, emphasizing unexpected words with emphatic stress.
Disgusted	**Voice:** curled-lipped and repulsed, with nasal resonance. **Tone:** scornful, dismissive, and revolted. **Pacing:** slightly drawn-out on offensive words, with sharp cutoffs elsewhere. **Emotion:** strong aversion, as if the subject is physically distasteful.
Excited	**Voice:** thrilled, animated, and high-energy, with rising intonation. **Tone:** enthusiastic and eager, barely able to contain yourself. **Pacing:** fast and breathless, tumbling from one idea to the next. **Emotion:** exuberant anticipation and joyful hype.
Sleepy	**Voice:** drowsy, soft, and slightly slurred. **Tone:** lethargic, relaxed, and half-awake. **Pacing:** slow and drawn out, with yawning pauses. **Emotion:** deep tiredness, as if you are about to drift off mid-sentence.
Sarcastic	**Voice:** dry, flat, and ironic, with pointed emphasis. **Tone:** mocking and knowing, clearly meaning the opposite of the words. **Pacing:** deliberate, with exaggerated stress on key words. **Emotion:** wry amusement and condescending disbelief.
Whispering	**Voice:** quiet, breathy whisper with almost no vocal cord engagement. **Tone:** intimate, secretive, and conspiratorial. **Pacing:** careful and deliberate, with clear articulation despite the low volume. **Emotion:** hushed urgency, as if sharing a secret that must not be overheard.

Table 6: Synthesis prompts for ten speaking styles, describing voice, tone, pacing, and emotion.

specified values, but query-matching requirements must identify the correct attribute. For example, accent matching must refer to "accent" rather than "speaking style" or "way of speaking."

Deduplication. Within each pattern of requirements, we consider instructions in random order and discard any instruction with a ROUGE-L F1 score of at least 0.7 against a previously retained instruction.

A.6 Statistics

The dataset contains 899,931 samples, comprising 800,000 from real data and 99,931 from synthetic data. Table 9 reports the training and validation counts by source.

Data Split. We shuffle the samples and keep those sharing the same spoken query in a single split. The resulting training and validation sets contain 890,278 and 9,653 samples, respectively.

Preventing Data Contamination. Our retrieval training data exclude test recordings from INSPIRE (Li and Lee, 2026) and the 15 evaluated MAEB (Assadi et al., 2026) tasks. We exclude MAEB tasks derived from CREMA-D and VoxCeleb because these corpora are included in our training data. Synthetic training uses MS MARCO text and UrbanSound8K sounds, whereas INSPIRE's Synthetic subset uses Natural Questions text and ESC-50 sounds. The synthesis voices, text-to-speech models, and instruction-generation models also differ from those used in INSPIRE.

Short instructions	Full-sentence instructions
You are generating {N} very short, informal retrieval cues for an audio retrieval system. Given a spoken query, the user wants to find another clip satisfying certain requirements. MATCHING constraints (same as spoken query): - the same {key} (e.g., {values}) as the given spoken query TARGET constraints (find a specific value): - a specific {key} (e.g., {values}) of "{value}" Each cue should be an informal, telegraphic phrase a user might type in a chat or voice interface: keyword-style fragments, not full sentences. Requirements: 1. Very short, typically 2 to 6 words 2. Lowercase is fine, no trailing punctuation 3. No full sentences; use fragments, comma-separated keywords, or elliptical phrases 4. For MATCH constraints use phrases such as "same emotion", "matching accent" 5. For TARGET constraints use phrases such as "female", "happy", "british accent" 6. A natural example for this pattern is "{example_cue}" 7. Vary wording, word order, and synonyms across cues; do not repeat the example 8. Do NOT mention technical terms such as "requirements", "attribute", or "metadata" Output ONLY the cues, one per line, numbered 1 to {N}.	You are generating diverse natural language retrieval instructions for an audio retrieval system. Given a spoken query, the user wants to find another clip satisfying certain requirements. MATCHING constraints (same as spoken query): - the same {key} (e.g., {values}) as the given spoken query TARGET constraints (find a specific value): - a specific {key} (e.g., {values}) of "{value}" Generate {N} diverse and natural retrieval instructions. Each instruction should: 1. Clearly indicate which attributes should MATCH the spoken query (e.g., "same emotion", "matching gender", "from the same speaker") 2. Clearly indicate which attributes should have a SPECIFIC TARGET value (e.g., "with a happy emotion", "in a British accent") 3. Use varied sentence structures, synonyms, and phrasings 4. Be natural and conversational 5. Not mention technical terms such as "requirements", "attribute", or "metadata" 6. Refer to "the spoken query" or "the given audio" when referring to the query Output ONLY the instructions, one per line, numbered 1 to {N}.

Table 7: Prompt templates for generating short instructions and full-sentence instructions. Only terminology is adapted to match the paper. The templates distinguish query-matching requirements in r_q from specified values in r_s. Braces mark fields filled for each set of requirements, and N denotes the requested number of instructions.

Requirements	Short instruction	Full-sentence instruction
Same speaker	identical voice	Fetch another clip of the speaker from this audio clip.
Same environmental sound	matching ambiance	Find audio that matches the background sound environment of this recording.
Same speaker, angry speaking style	anger, same voice	Search for an angry version of this person's voice.

Table 8: Examples of short instructions and full-sentence instructions for the same retrieval requirements. "Same" indicates a query-matching requirement in r_q. The final row also includes a specified speaking style of "angry" in r_s.

Source	Training	Validation
Real data		
VoxCeleb1 (Nagrani et al., 2017)	294,303	2,974
ESD (Zhou et al., 2021)	115,282	1,182
MELD (Poria et al., 2019)	72,526	791
CREMA-D (Cao et al., 2014)	56,754	584
L2-ARCTIC (Zhao et al., 2018)	53,022	537
LibriSpeech-C (Lin et al., 2024b)	51,553	521
EmoV-DB (Adigwe et al., 2018)	49,834	525
MSP-IMPROV (Busso et al., 2016)	44,763	459
VocalSound (Gong et al., 2022)	20,678	209
SpokenWOZ (Si et al., 2023)	13,361	135
MultiDialog (Park et al., 2024)	12,941	131
UrbanSound8K (Salamon et al., 2014)	5,844	59
EMNS (Noriy et al., 2023)	1,022	10
Real data subtotal	791,883	8,117
Synthetic data		
MS MARCO (Nguyen et al., 2016)	98,395	1,536
Total	**890,278**	**9,653**

Table 9: Training and validation sample counts by source. Each sample comprises a spoken query, an instruction, and positive and negative spoken documents.

B Training Details

B.1 Setup

Setting	Value
Learning rate	2×10^{-4}
Adam coefficients	$\beta_1 = 0.9, \beta_2 = 0.95$
Adam epsilon	10^{-8}
Weight decay	0.1
Maximum gradient norm	1.0
Schedule	Linear warmup, then cosine decay to zero
Warmup	First 5% of optimizer steps
Training duration	2 epochs; 3,476 optimizer steps
Global batch size	512 samples
Batch size per GPU	32 samples
Contrastive temperature	0.03

Table 10: Training settings shared by both models. The global batch contains 32 samples on each of 16 GPUs.

We train IRIS-Embed using Qwen2.5-Omni-3B (Xu et al., 2025a)[9] and Gemma-4-E2B-it (Team et al., 2026)[10] with the ms-swift package (Zhao et al., 2024)[11]. We optimize the contrastive objective in Section 3.3 using the data splits in Appendix A.6. We evaluate the final checkpoint after two epochs.

Optimization. Both models use fused AdamW with the same learning rate for all trainable components. Weight decay is not applied to biases or normalization parameters. Training uses bfloat16 precision and DeepSpeed ZeRO stage 2 across 16 NVIDIA GB200 GPUs. Table 10 summarizes the optimization settings.

Trainable Components. The audio layer weighting module g uses the pretrained `all-MiniLM-L6-v2` sentence encoder (Wang et al., 2020; Reimers and Gurevych, 2019)[12]. Both the audio encoder and sentence encoder remain frozen. We adapt the language model using LoRA with rank 64, scaling parameter 128, and dropout 0.05. The audio projection and the added normalization and linear layers in g are trained directly alongside the LoRA parameters.

LoRA Configuration. For Qwen2.5-Omni-3B, we apply LoRA to `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj` across all 36 language-model layers. For Gemma-4-E2B-it, we adapt the applicable attention and feed-forward projections across all 35 language-model layers, along with `per_layer_input_gate`, `per_layer_projection`, and the shared `per_layer_model_projection`. Since later layers reuse keys and values, LoRA adapters for `k_proj` and `v_proj` are applied only to the first 15 layers.

Data Ordering. Before training, we arrange samples by source and spoken query in groups of approximately 64. Grouping by source emphasizes within-source negatives, particularly for smaller datasets. For dialogue continuation, this ordering is intended to reduce shortcuts based on speaker identity or recording conditions. Grouping by spoken query places different instructions for the same query close together. Different instructions may select different positive documents, so this ordering is intended to encourage retrieval to depend on the instruction's requirements.

B.2 Audio Layer Weighting

Computation. To compute $g(p)$, we mean-pool MiniLM's non-padding token hidden states, then apply learned layer normalization and a linear layer. We apply softmax to obtain weights for the pre-encoder hidden states and all audio-encoder layer hidden states: 33 hidden-state sequences in Qwen2.5-Omni-3B and 13 in Gemma-4-E2B-it. Each model pools and projects the weighted audio hidden states using its existing operations, including normalization, before passing them to the language model.

Initialization. We initialize the linear weights from a Gaussian distribution with mean 0 and standard deviation 10^{-4}. For Qwen2.5-Omni-3B, all biases are initialized to zero, yielding approximately uniform layer weights. For Gemma-4-E2B-it, we initialize the layer weights to emphasize the final audio layer, reducing the initial deviation from its original representation. Let N denote the number of audio hidden-state sequences, including the pre-encoder hidden states. We set the final-layer bias to

$$b_{\text{final}} = \log \frac{0.9(N-1)}{0.1}, \tag{1}$$

while initializing all other biases to zero. For Gemma-4-E2B-it, $N = 13$, so this initialization assigns approximately 90% of the initial weight to

[9] https://huggingface.co/Qwen/Qwen2.5-Omni-3B
[10] https://huggingface.co/google/gemma-4-E2B-it
[11] https://github.com/modelscope/ms-swift
[12] https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

Qwen2.5-Omni-3B Query Prompt:	**Qwen2.5-Omni-3B Document Prompt:**
<|im_start|>user Instruction: [instruction] Query: <|audio_bos|>[audio_tokens]<|audio_eos|> <|im_end|> <|im_start|>assistant	<|im_start|>user Document: <|audio_bos|>[audio_tokens] <|audio_eos|><|im_end|> <|im_start|>assistant
Gemma-4-E2B-it Query Prompt:	**Gemma-4-E2B-it Document Prompt:**
<bos><|turn>user Instruction: [instruction] Query: [audio_tokens]<turn|> <|turn>model	<bos><|turn>user Document: [audio_tokens]<turn|> <|turn>model
MiniLM Query Prompt:	**MiniLM Document Prompt:**
Instruction: [instruction] Query:	Document:

Table 11: IRIS-Embed prompt templates. Red: input labels; blue: placeholders.

the final layer and distributes the remainder across other layers.

B.3 Prompts

Table 11 presents our prompt templates, adapted from Qwen3-VL-Embedding (Li et al., 2026a) to each model's chat and audio formats. The query prompt includes the instruction and spoken query, while the document prompt contains the spoken document. We extract the final-layer hidden state of the last non-padding input token as the embedding without generating a response.

For audio layer weighting, MiniLM processes text-only prompts. The query prompt includes the instruction, while the document prompt remains fixed across all spoken documents.

C Evaluation Details

C.1 INSPIRE

Benchmark. INSPIRE (Li and Lee, 2026)[13] evaluates spoken-document retrieval given a spoken query and a natural-language instruction. Instructions describe requirements for speech attributes. Thus, the same spoken query may have different relevant documents depending on the instruction. A document is relevant only if it satisfies all requirements.

Tasks. DailyTalk (Lee et al., 2023) uses the first half of a dialogue as the spoken query and evaluates retrieval of its continuation. VCTK (Yamagishi et al., 2019) evaluates speaker matching using hard negatives with the same spoken content but different speakers. Expresso (Nguyen et al., 2023) evaluates speaker and speaking-style matching, separately or jointly, including instructions with specified speaking styles.

The Synthetic subset combines spoken queries and documents synthesized from Natural Questions (Kwiatkowski et al., 2019) using GPT-4o mini TTS (OpenAI, 2024) with environmental sounds from ESC-50 (Piczak, 2015). Synthetic evaluates answer retrieval and matching by speaker, speaking style, and environmental sound, including instructions with multiple attributes. Speaking style and environmental sound may match the spoken query or take values specified in the instruction. Appendix A.6 describes the separation between training and evaluation data.

Metric. For each query–instruction pair, embedding methods rank documents in the corresponding subset by cosine similarity. Recall@K measures the fraction of relevant documents retrieved among the top K results. Let Q_Z denote the evaluated query–instruction pairs, $\pi_K(q, z)$ the top K retrieved documents, and $D^+(q, z)$ the relevant documents. We compute average recall as

$$\text{Recall@}K = \frac{1}{|Q_Z|} \sum_{(q,z)\in Q_Z} \frac{|\pi_K(q, z) \cap D^+(q, z)|}{|D^+(q, z)|}. \quad (2)$$

Each query–instruction pair contributes equally to the average. We report Recall@K as percentages for $K \in \{10, 50, 100\}$ on each subset. For the instruction-type analysis in Section 6.1, we compute Recall@50 separately for each instruction type.

C.2 Comparison Settings

Table 1 includes published baseline results.

Random. The random baseline averages uniform random document rankings over 100 runs (Li and Lee, 2026).

General Speech Encoders. WavLM-large (Chen et al., 2022)[14] encodes spoken queries and documents without instructions. Embeddings are obtained by mean-pooling its final-layer hidden states over time.

Specialized Speech Encoders. ECAPA-TDNN (Desplanques et al., 2020)[15] and emotion2vec-base (Ma et al., 2024)[16] extract speaker and emotion embeddings from their final-layer representations, respectively. Both encode spoken queries and documents without instructions.

Cascaded Retrieval. Whisper-large-v3 (Radford et al., 2023)[17] generates transcripts, and Qwen3-Omni-30B-A3B-Captioner (Xu et al., 2025b)[18] generates captions for spoken queries and documents. Qwen3-Embedding-8B (Zhang et al., 2025b)[19] encodes the transcripts and captions, with the instruction included in the query input.

LALM Embeddings. Qwen2.5-Omni-3B (Xu et al., 2025a)[20], Gemma-4-E2B-it (Team et al., 2026)[21], and Qwen3-Omni-30B (Xu et al., 2025b)[22] follow INSPIRE's evaluation setup (Li and Lee, 2026) without additional embedding training. For these models, query and document inputs end with "Summarize the above sentence and speech in one word" and "Summarize the

[13] https://huggingface.co/datasets/lca0503/INSPIRE

[14] https://huggingface.co/microsoft/wavlm-large

[15] https://huggingface.co/speechbrain/spkrec-ecapa-voxceleb

[16] https://huggingface.co/emotion2vec/emotion2vec_base

[17] https://huggingface.co/openai/whisper-large-v3

[18] https://huggingface.co/Qwen/Qwen3-Omni-30B-A3B-Captioner

[19] https://huggingface.co/Qwen/Qwen3-Embedding-8B

[20] https://huggingface.co/Qwen/Qwen2.5-Omni-3B

[21] https://huggingface.co/google/gemma-4-E2B-it

[22] https://huggingface.co/Qwen/Qwen3-Omni-30B-A3B-Instruct

Task	Type	Instruction
BeijingOpera	Clf	Find recordings with the same percussion instrument as the query.
BirdCLEF	Clf	Find recordings with the same bird species as the query.
CommonLanguageAgeDetection	Clf	Find recordings with the same speaker age group as the query.
GTZANGenre	Clf	Find recordings with the same music genre as the query.
IEMOCAPGender	Clf	Find recordings with the same gender as the query.
MInDS14	Clf	Find recordings with the same intent as the query.
MridinghamTonic	Clf	Find recordings with the same tonic as the query.
SIBFLEURS	Clf	Find recordings with the same topic as the query.
VoxPopuliLanguageID	Clf	Find recordings with the same spoken language as the query.
VehicleSoundClustering	Clust	Find recordings with the same environmental sound as the query.
VoxPopuliGenderClustering	Clust	Find recordings with the same gender as the query.
NMSQAPairClassification	PC	Find recordings that contain the answer to the query.
VoxPopuliAccentPairClassification	PC	Find recordings with the same accent as the query.
GTZANAudioReranking	Rrnk	Find recordings with the same music genre as the query.
JamAltArtistA2ARetrieval	Rtrvl	Find recordings with the same artist as the query.

Table 12: Task instructions used to extract IRIS-Embed query embeddings for MAEB evaluation. Clf: classification; Clust: clustering; PC: pair classification; Rrnk: reranking; Rtrvl: retrieval.

above speech in one word", respectively. LCO-Embedding-Omni-3B (Xiao et al., 2025)[23] follows its published embedding extraction configuration. All four methods extract embeddings from the final-layer hidden state of the last non-padding input token without generating a response.

IRIS-Embed. We evaluate the final IRIS-Embed checkpoints from Appendix B with learned audio layer weighting. Table 11 gives the query and document prompt templates. We extract the final-layer hidden state of the last non-padding input token as the embedding without generating a response.

C.3 Layer Aggregation

Table 2 compares audio layer aggregation for Qwen2.5-Omni-3B. Layer 13 and Final Layer use the 13th and final audio-encoder hidden states, respectively, followed by pooling and projection into the language model. Mean Layer assigns equal weights to the pre-encoder hidden states and all layer hidden states. Finetuned updates the audio encoder during retrieval training, while the other variants keep it frozen. Static Weight learns a single set of layer weights shared across all queries and documents, independent of instructions or prompts. Inst. Weight is our instruction-aware audio layer weighting method, which predicts weights from text prompts as described in Section 3.2.1. Query weights vary with the instruction, while document weights are computed from a fixed prompt. Mean Layer, Static Weight, and Inst. Weight combine layers 0–32 for Qwen2.5-Omni-3B, with layer 0 denoting the pre-encoder hidden states. Both weighting variants use the same instructions in the language-model query prompt.

Final Layer, Mean Layer, Static Weight, and Inst. Weight use the training data and language-model LoRA settings in Appendix B and train the audio projection. For Gemma-4-E2B-it, we compare Final Layer, Mean Layer, and Inst. Weight. Appendix D.3 provides the full results for both models. Appendix D.4 separately examines frozen and trainable audio projections for Final Layer.

C.4 MAEB

Tasks. We evaluate 15 audio tasks from MAEB (Assadi et al., 2026)[24], following the task definitions and metrics of its audio-only beta leaderboard.[25] We exclude tasks derived from CREMA-D (Cao et al., 2014) and VoxCeleb (Nagrani et al., 2017), as these corpora are included in our retrieval training data. Table 3 reports scores on a 0–100 scale using shortened task names.

Baseline Scores. We evaluate the Qwen2.5-Omni-3B and Gemma-4-E2B-it baselines with the same task instructions as IRIS-Embed. Scores for the other baselines come from the MAEB beta leaderboard and are averaged uniformly across language subsets for each task.

[23] https://huggingface.co/LCO-Embedding/LCO-Embedding-Omni-3B

[24] https://github.com/embeddings-benchmark/mteb

[25] https://mteb-leaderboard.hf.space/benchmark/MAEB(beta)

Embedding Extraction. Table 12 lists the full task names and corresponding instructions. For all tasks except retrieval, we encode every recording with its task instruction using the query prompt in Table 11. Retrieval uses separate query and document prompts, as described below. Each input passes through the full IRIS-Embed model, and we extract the final-layer hidden state of the last non-padding input token as its embedding. Model parameters remain fixed throughout evaluation.

Classification. Following MAEB (Assadi et al., 2026), nine tasks evaluate classification of percussion instruments (Tian et al., 2014), bird species (Klinck et al., 2025), speaker age groups (Sinisetty et al., 2021), music genres (Tzanetakis and Cook, 2002), gender (Busso et al., 2008), intent (Gerz et al., 2021), musical tonic (Anantapadmanabhan et al., 2013), topic (Schmidt et al., 2025), and language (Wang et al., 2021). Evaluation uses a logistic-regression probe fitted on fixed embeddings with eight samples per class. Accuracy is averaged over ten resamples.

Clustering. VehicleSound (Bazilinskyy et al., 2018) and VoxPopuliGender (Wang et al., 2021) evaluate clustering by sound class and gender, respectively. Following MAEB, MiniBatchKMeans uses the number of reference classes as the number of clusters. Performance is measured by V-measure (Rosenberg and Hirschberg, 2007) over bootstrapped subsamples.

Pair Classification. NMSQA (Lin et al., 2022) and VoxPopuliAccent (Wang et al., 2021) evaluate the relevance of recording pairs. Following MAEB, we report the highest average precision across the benchmark's similarity measures.

Reranking. GTZAN (Tzanetakis and Cook, 2002) evaluates reranking of a provided set of relevant documents and hard negatives using cosine similarity. We report mean average precision at 1,000, denoted MAP@1000.

Retrieval. JamAlt (Cífka et al., 2024) evaluates audio-to-audio retrieval of songs by the same artist as the query. The query prompt includes the query recording and task instruction, while the document prompt contains the candidate recording without an instruction. We report normalized discounted cumulative gain at 10, denoted nDCG@10.

D Full Results

D.1 Instruction Types

Table 13 reports Recall@50 by instruction type on INSPIRE, covering instructions with a single attribute or multiple attributes. Relative to the corresponding LALM embedding baselines, IRIS-Embed improves Recall@50 in all 21 evaluation settings for Qwen2.5-Omni-3B and 20 of 21 for Gemma-4-E2B-it. The exception is matching speaking style with a specified environmental sound on Synthetic, where Gemma-4-E2B-it achieves 1.25% with IRIS-Embed versus 1.50% with the LALM baseline.

IRIS-Embed also outperforms other baselines across several instruction types. Both models outperform all LALM embedding baselines on speaker matching in VCTK and all four instruction types in Expresso. Both models also outperform all comparison baselines on speaking-style matching in Expresso and Synthetic, and environmental-sound matching in Synthetic.

For instructions with multiple attributes, the Qwen2.5-Omni-3B and Gemma-4-E2B-it versions of IRIS-Embed outperform all comparison baselines in 11 and 10 of the 13 evaluation settings across Expresso and Synthetic, respectively. On Synthetic, matching the spoken query's speaker, speaking style, and environmental sound together gives Recall@50 of 61.75% and 31.50%, compared with 20.50% for the strongest baseline, ECAPA-TDNN. Matching the speaker and a specified speaking style yields 46.00% and 45.25%, compared with 2.00% for the strongest baseline. The gains therefore extend to requirements with query-matched and explicitly specified attribute values.

D.2 Instruction Awareness

We assess whether IRIS-Embed follows retrieval instructions by varying the instruction while keeping the spoken queries, document collection, and relevance labels fixed for each set of evaluation requirements. The tables report Recall@50, with instructions in rows and evaluation requirements in columns. Tables 1 and 13 use the original INSPIRE instructions. Here, we replace them with fixed short or full-sentence instructions, so recall may differ from those results.

We consider five instructions with a single attribute: dialogue continuation, answer retrieval, speaker matching, speaking-style matching, and environmental-sound matching. Dialogue continuation and answer retrieval specify different relevance requirements for spoken content. Table 14 lists a short instruction and a full-sentence instruction for each of the five instruction types. The left sides of Figures 5 and 7 show the Short rows from Table 15 for Qwen2.5-Omni-3B and Gemma-4-E2B-it, respectively. For both models, the instruction matching the evaluation requirements achieves the highest Recall@50 across all four Synthetic evaluation settings.

Table 15 reports both wording forms, labeled Short and Full, and a no-instruction condition. Within each wording form, the matching instruction achieves the highest Recall@50 for both models across all four Synthetic evaluation settings and for speaker and speaking-style matching on Expresso. The matching instruction also performs best for Qwen2.5-Omni-3B on VCTK and Gemma-4-E2B-it on DailyTalk. Both short instructions and full-sentence instructions therefore guide retrieval according to the requested requirements in most evaluation settings.

The matching instruction does not always achieve the highest recall. On DailyTalk, Qwen2.5-Omni-3B performs best with the answer-retrieval instruction under both wording forms. This may reflect overlap between answering a question and continuing a dialogue. On VCTK, Gemma-4-E2B-it achieves 76.25% with the short continuation instruction and 75.00% with the short speaker-matching instruction. The corresponding full-sentence instructions and no instruction all yield 75.63%, suggesting that speaker information remains available without an explicit speaker-matching instruction.

We further evaluate four short instructions with multiple attributes of the spoken query. These cover all pairwise combinations of speaker, speaking style, and environmental sound, as well as all three attributes together. The right panels of Figures 5 and 7 show the rows for these four short instructions in Table 16 across the corresponding Synthetic evaluation settings. The matching instruction achieves the highest Recall@50 in all four evaluation settings for Qwen2.5-Omni-3B and three of four for Gemma-4-E2B-it. When relevance requires matching all three attributes, the corresponding instruction outperforms all pairwise instructions for both models.

Table 16 provides the full comparison, including the five short instructions above and no instruction. For both models, the matching instruction outper-

Method	Model	DailyTalk	VCTK	Expresso				Synthetic			
		Cont.	Spk.	Spk.	Sty.	Spk. +Sty.	Spk. +Sty.*	Ans.	Spk.	Sty.	Env.
RAND	Random	1.06	1.84	1.73	1.46	1.38	1.42	0.84	1.09	1.38	1.04
ENC	WavLM-large	24.50	35.00	11.50	30.75	27.50	1.00	0.75	0.75	8.75	7.50
SENC	ECAPA-TDNN	24.00	**100.00**	**71.50**	24.25	**70.50**	0.75	1.67	**87.75**	13.50	1.75
	emotion2vec-base	23.00	6.25	3.25	6.25	7.25	0.75	16.75	2.25	18.75	2.25
CAS	Qwen3-Embedding-8B	**81.50**	6.25	4.00	5.50	6.00	6.75	**98.50**	1.75	1.00	4.50
LALM	Qwen2.5-Omni-3B	53.00	3.13	2.00	6.00	3.75	3.25	64.75	1.00	1.25	0.50
	Gemma-4-E2B-it	25.50	4.38	1.50	1.25	0.75	1.25	50.92	0.50	1.50	0.50
	Qwen3-Omni-30B	74.50	3.13	2.00	5.25	4.50	2.50	80.75	1.25	2.00	2.00
	LCO-Embedding-Omni-3B	75.50	2.50	2.25	16.25	8.25	7.00	97.25	1.25	1.75	6.00
IRIS	Qwen2.5-Omni-3B	77.50	85.00	41.25	42.00	62.00	**34.75**	90.50	53.50	**53.75**	**35.50**
	Gemma-4-E2B-it	44.50	74.38	33.00	**48.00**	63.75	30.50	80.58	51.25	47.25	8.50

Method	Model	Synthetic										
		Spk. +Sty.	Spk. +Sty.*	Spk. +Env.	Spk. +Env.*	Sty. +Env.	Sty. +Env.*	Sty.* +Env.	Spk. +Sty. +Env.	Spk. +Sty.* +Env.	Spk. +Sty. +Env.*	Spk. +Sty.* +Env.*
RAND	Random	1.04	1.04	0.96	0.96	0.90	0.96	0.96	0.92	0.98	0.98	0.91
ENC	WavLM-large	6.25	1.50	7.25	0.75	9.00	1.00	3.25	12.25	5.50	2.25	0.50
SENC	ECAPA-TDNN	**61.50**	2.00	7.00	4.50	2.75	3.75	0.00	20.50	1.25	13.50	1.00
	emotion2vec-base	2.75	1.75	4.00	1.25	5.50	0.50	1.25	6.00	2.25	1.75	1.00
CAS	Qwen3-Embedding-8B	2.50	1.50	3.00	2.75	2.00	2.75	3.25	4.25	1.75	2.25	2.50
LALM	Qwen2.5-Omni-3B	1.50	0.50	0.50	0.75	2.00	1.25	0.25	0.50	0.75	1.50	1.00
	Gemma-4-E2B-it	1.75	1.50	1.00	2.00	1.75	1.50	1.50	0.75	1.25	0.75	1.25
	Qwen3-Omni-30B	1.00	0.50	0.50	0.75	1.25	0.75	0.75	1.00	1.25	0.75	1.25
	LCO-Embedding-Omni-3B	1.75	1.00	2.75	3.00	6.25	3.00	2.50	4.50	2.75	4.25	2.50
IRIS	Qwen2.5-Omni-3B	54.25	**46.00**	**47.50**	**17.25**	**35.75**	**15.00**	**38.00**	**61.75**	**57.00**	**30.00**	**26.75**
	Gemma-4-E2B-it	47.00	45.25	15.50	7.25	13.50	1.25	8.25	31.50	18.25	15.00	13.75

Table 13: Full Recall@50 results by instruction type on INSPIRE. Bold and underline indicate the best and second-best results, respectively. Cont.: dialogue continuation; Ans.: answer retrieval; Spk.: speaker; Sty.: speaking style; Env.: environmental sound. + combines requirements; ∗ marks a specified value.

forms these additional conditions in all four evaluation settings. These findings show that IRIS-Embed responds to instructions with multiple attributes.

D.3 Layer Aggregation

Table 17 reports Recall@10, Recall@50, and Recall@100 for audio layer aggregation, while Table 18 provides Recall@50 by instruction type. For Qwen2.5-Omni-3B, the comparison includes Layer 13, Final Layer, Mean Layer, Finetuned, Static Weight, and Inst. Weight. For Gemma-4-E2B-it, we compare Final Layer, Mean Layer, and our method. Appendix C.3 describes these methods and their training settings.

For Qwen2.5-Omni-3B, our method achieves the highest macro-average Recall@50 of 62.92% in Table 2, a gain of 14.36 percentage points over Final Layer. It consistently outperforms Final Layer on VCTK, Expresso, and Synthetic. Final Layer remains higher on DailyTalk at Recall@50.

Layer 13 and Finetuned improve speaker and speaking-style matching over Final Layer, but have much lower dialogue-continuation recall. Finetuned consistently leads on VCTK and Layer 13 on Expresso. Our method consistently outperforms both variants on DailyTalk and Synthetic. These methods trade off performance across retrieval requirements.

Our method also exceeds Mean Layer and Static Weight in Recall@50 on DailyTalk, VCTK, and Expresso. The macro-average Recall@50 gains are 5.60 and 4.61 percentage points, respectively. Both alternatives remain higher on Synthetic at Recall@50. Static Weight also exceeds our method on Synthetic at Recall@10 and Recall@100, and on Expresso at Recall@100.

The instruction-type results further distinguish these methods. On answer retrieval, our method achieves 90.50% Recall@50, compared with 68.25% for Finetuned and 2.00% for Layer 13. In the pairwise comparison with Static Weight, our method has higher Recall@50 in seven of eight evaluation settings with a single attribute. For instructions with multiple attributes, Static Weight

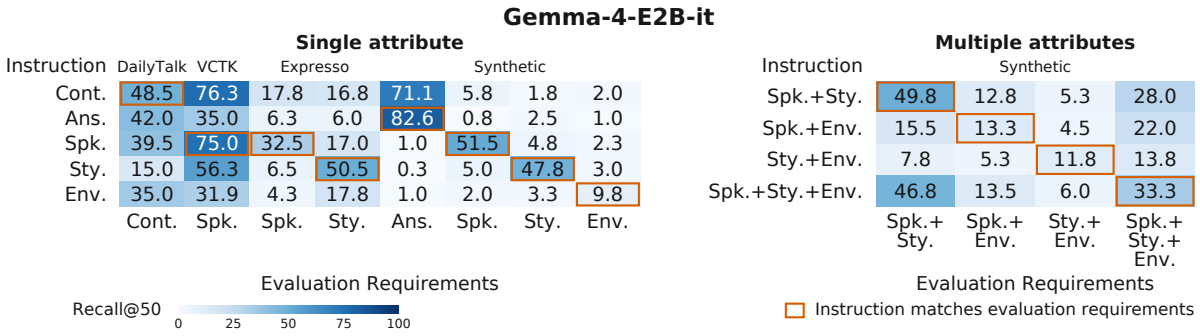

Figure 7: Recall@50 for IRIS-Embed with Gemma-4-E2B-it. Cont./Ans.: dialogue continuation/answer retrieval; Spk./Sty./Env.: speaker/speaking style/environmental sound; +: multiple attributes.

Label	Short instruction	Full-sentence instruction
Cont.	Continuation.	Find speech that continues the conversation in the query.
Ans.	Find the answer.	Find speech that contains the answer to the question in the query.
Spk.	Same speaker.	Find speech from the same speaker as the query.
Sty.	Same speaking style.	Find speech with the same speaking style as the query.
Env.	Same environmental sound.	Find speech with the same environmental sound as the query.
Doc.	`Document:`	

Table 14: Instructions for instruction awareness and layer-weight analysis. The main-paper instruction-awareness and layer-weight analyses use short instructions. Doc. denotes the document prompt used in the layer-weight analysis.

has higher recall in eight of thirteen settings, our method in four, and one is tied.

With Gemma-4-E2B-it, our method consistently outperforms Final Layer on VCTK, Expresso, and Synthetic, while Mean Layer leads on Expresso. Answer retrieval Recall@50 increases from 1.50% with Mean Layer to 80.58% with our method. However, Mean Layer performs better on several instructions that combine speaker matching with other attributes. The aggregate gains support the effectiveness of instruction-aware audio layer weighting with a frozen audio encoder.

D.4 Audio Projection

We examine whether training the audio projection improves retrieval using the final audio-encoder hidden states. Both variants freeze the audio encoder and train the language-model LoRA parameters. Frozen keeps the audio projection fixed, while Trainable updates it during retrieval training. Tables 19 and 20 report results by subset and instruction type, respectively.

For both models, the trainable audio projection consistently outperforms the frozen projection on VCTK, Expresso, and Synthetic. On Synthetic, these gains cover all eleven instruction types with multiple attributes at Recall@50. The effects on spoken-content retrieval vary between models. At Recall@50, Qwen2.5-Omni-3B improves on answer retrieval but declines on dialogue continuation, while Gemma-4-E2B-it shows the opposite trend.

With a trainable audio projection in both settings, Weighted Layer consistently outperforms Final Layer on VCTK, Expresso, and Synthetic for both models. Thus, the observed gains over Final Layer persist when both variants train the audio projection.

D.5 Layer Weights

We analyze how audio layer weights vary across retrieval requirements and instruction wording using the ten instructions in Table 14. The document prompt serves as a reference without an instruction. All prompts follow the MiniLM templates in Table 11. Figure 8 presents all eleven prompts for both models, while Figure 6 focuses on the five short instructions and the document prompt for Qwen2.5-Omni-3B. For Gemma-4-E2B-it, layer 4 receives the highest adjusted weight under every query instruction except the short answer-retrieval instruction, which places the highest weight on the final layer.

To account for differences in hidden-state scale, we adjust the layer weights using the same 680 unique INSPIRE spoken queries for every prompt

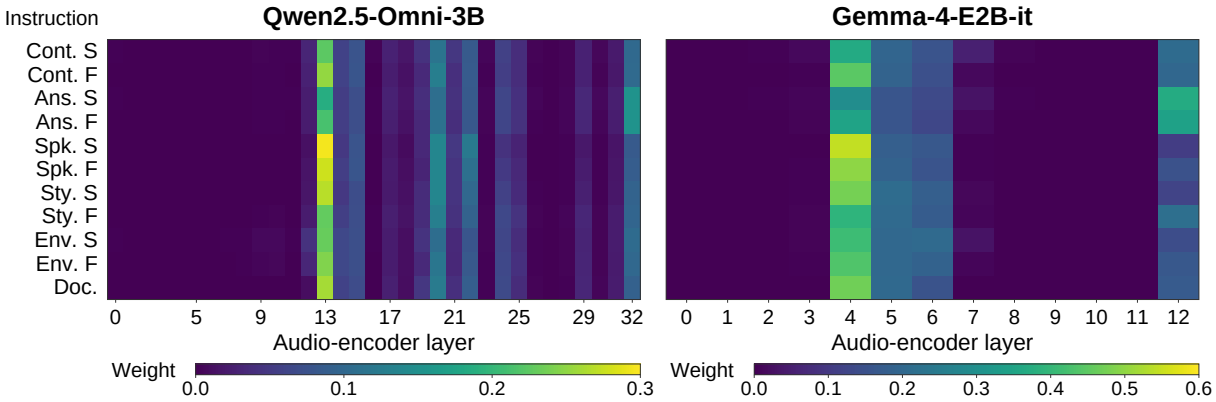

Figure 8: Audio layer weights adjusted for hidden-state scale. S/F: short/full-sentence instructions; Doc.: document prompt; 0: pre-encoder hidden states. Encoder layers are indexed from one. Color scales differ between models.

and both models. Let $w_\ell(p)$ denote the original weight for layer ℓ under text prompt p. For a recording x, let $s_\ell(x)$ denote the root mean square of the hidden states across hidden dimensions, averaged over valid frames. We compute the adjusted weights as

$$\widehat{w}_\ell(x,p) = \frac{w_\ell(p)s_\ell(x)}{\sum_k w_k(p)s_k(x)},$$

where k and ℓ index the pre-encoder hidden states and all audio-encoder layer hidden states. The adjusted weights sum to one for each recording and prompt. We first average them within each INSPIRE subset, then average the four subset means equally. This adjustment is used only for analysis and does not affect model computation. The adjusted weights describe the relative scales of weighted audio layer hidden states, not their causal contributions to retrieval scores.

D.6 Layer Representations

We extend the analysis in Section 3.2.1 to examine whether audio projection affects the best-performing layer. For each layer, we apply the same frozen audio projection, mean-pool the projected hidden states over time, and rank spoken documents by cosine similarity. Both the audio encoder and projection remain frozen, and the language model is not used. Figure 9 reports Recall@50 for dialogue continuation on DailyTalk, speaker matching on VCTK, speaking-style matching on Expresso, and answer retrieval on Synthetic. The best-performing layer continues to vary across retrieval requirements after projection.

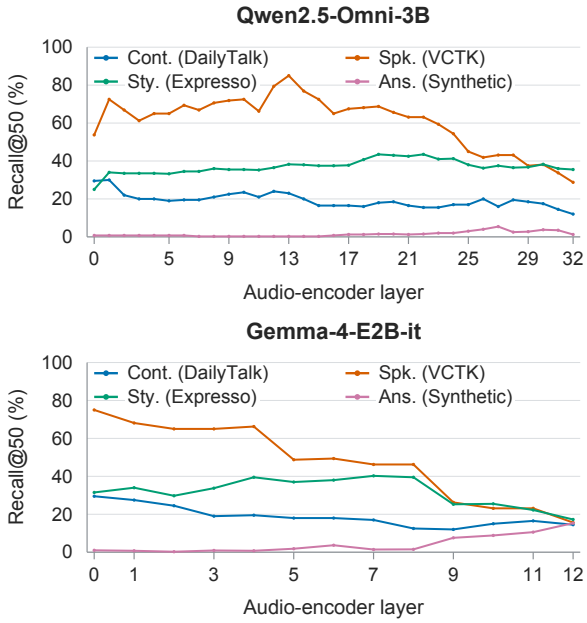

Figure 9: Recall@50 using frozen audio-encoder hidden states after audio projection. 0 denotes the pre-encoder hidden states; encoder layers are indexed from one.

Qwen2.5-Omni-3B									
Instruction	**Form**	**DailyTalk**	**VCTK**	**Expresso**		**Synthetic**			
		Cont.	**Spk.**	**Spk.**	**Sty.**	**Ans.**	**Spk.**	**Sty.**	**Env.**
Continuation.	Short	79.00	81.88	19.75	22.50	87.25	2.75	2.75	2.75
Find speech that continues the conversation in the query.	Full	77.50	83.75	31.00	15.75	63.00	18.75	2.50	2.00
Find the answer.	Short	82.50	73.75	8.00	4.25	90.25	2.00	1.75	1.25
Find speech that contains the answer to the question in the query.	Full	78.50	67.50	6.25	3.50	91.00	2.00	1.50	0.25
Same speaker.	Short	63.50	85.63	40.75	13.50	1.50	53.00	5.50	0.75
Find speech from the same speaker as the query.	Full	67.50	85.63	41.00	13.50	1.50	53.00	6.00	0.75
Same speaking style.	Short	10.50	30.00	4.00	42.50	0.50	2.75	55.00	0.50
Find speech with the same speaking style as the query.	Full	21.00	30.00	3.00	41.00	0.50	2.75	55.75	1.00
Same environmental sound.	Short	20.50	34.38	4.50	8.75	1.50	0.75	0.50	36.75
Find speech with the same environmental sound as the query.	Full	29.00	35.63	4.75	10.00	1.50	1.00	0.50	37.25
No instruction		75.00	80.63	20.75	26.00	74.58	3.25	3.75	8.75

Gemma-4-E2B-it									
Instruction	**Form**	**DailyTalk**	**VCTK**	**Expresso**		**Synthetic**			
		Cont.	**Spk.**	**Spk.**	**Sty.**	**Ans.**	**Spk.**	**Sty.**	**Env.**
Continuation.	Short	48.50	76.25	17.75	16.75	71.08	5.75	1.75	2.00
Find speech that continues the conversation in the query.	Full	49.00	75.63	19.25	13.75	73.83	3.50	1.25	1.25
Find the answer.	Short	42.00	35.00	6.25	6.00	82.58	0.75	2.50	1.00
Find speech that contains the answer to the question in the query.	Full	41.50	36.25	6.75	6.00	83.08	1.00	2.25	1.25
Same speaker.	Short	39.50	75.00	32.50	17.00	1.00	51.50	4.75	2.25
Find speech from the same speaker as the query.	Full	45.50	75.63	32.25	16.75	1.50	52.75	4.75	2.50
Same speaking style.	Short	15.00	56.25	6.50	50.50	0.25	5.00	47.75	3.00
Find speech with the same speaking style as the query.	Full	22.50	54.38	7.25	47.25	0.25	4.50	46.75	2.50
Same environmental sound.	Short	35.00	31.88	4.25	17.75	1.00	2.00	3.25	9.75
Find speech with the same environmental sound as the query.	Full	41.00	35.63	5.00	14.00	1.00	1.75	2.50	9.50
No instruction		44.00	75.63	20.00	39.00	8.75	21.25	24.25	3.75

Table 15: Recall@50 for IRIS-Embed with short instructions, full-sentence instructions, or no instruction. Rows show the exact instructions; Short/Full denote short/full-sentence instructions. Cont./Ans.: dialogue continuation/answer retrieval; Spk./Sty./Env.: speaker/speaking style/environmental sound.

Instruction	Qwen2.5-Omni-3B				Gemma-4-E2B-it			
	Spk. +Sty.	**Spk. +Env.**	**Sty. +Env.**	**Spk. +Sty. +Env.**	**Spk. +Sty.**	**Spk. +Env.**	**Sty. +Env.**	**Spk. +Sty. +Env.**
Find the answer.	0.75	1.75	3.25	2.00	1.00	1.25	1.75	1.50
Continuation.	1.50	4.50	5.25	6.50	2.75	3.75	3.25	4.50
Same speaker.	18.50	8.50	2.50	10.25	18.00	8.50	4.00	14.25
Same speaking style.	9.50	2.25	3.25	7.00	12.25	4.50	7.75	12.25
Same environmental sound.	1.50	10.75	15.00	11.75	2.50	6.75	9.25	10.50
Same speaker and same speaking style.	52.00	11.50	7.50	27.25	49.75	12.75	5.25	28.00
Same speaker and same environmental sound.	8.25	45.25	11.75	45.25	15.50	13.25	4.50	22.00
Same speaking style and same environmental sound.	6.00	11.50	35.00	36.75	7.75	5.25	11.75	13.75
Same speaker and same speaking style and same environmental sound.	30.00	29.25	22.25	65.00	46.75	13.50	6.00	33.25
No instruction	6.00	13.25	9.50	17.75	35.75	11.25	7.75	30.25

Table 16: Recall@50 for IRIS-Embed under short instructions and no instruction. Columns evaluate Synthetic requirements involving multiple attributes. Rows show the exact instructions; No instruction indicates an empty instruction. Spk., Sty., and Env. match the spoken query's speaker, speaking style, and environmental sound; + combines requirements.

Layer Aggregation	DailyTalk			VCTK			Expresso			Synthetic			Avg.		
	R@10	**R@50**	**R@100**	**R@10**	**R@50**	**R@100**	**R@10**	**R@50**	**R@100**	**R@10**	**R@50**	**R@100**	**R@10**	**R@50**	**R@100**
Qwen2.5-Omni-3B															
Layer 13	5.50	15.50	22.50	61.88	76.88	84.38	**32.44**	**48.94**	**57.75**	17.95	35.27	46.12	29.44	44.15	52.69
Final Layer	59.50	**82.50**	85.50	34.38	48.75	59.38	15.88	28.25	36.88	18.47	34.72	45.20	32.06	48.56	56.74
Mean Layer	47.50	70.00	78.50	57.50	71.25	81.25	25.75	43.13	52.63	26.03	44.90	55.00	39.20	57.32	66.85
Finetuned	14.50	32.50	42.00	**76.25**	**91.25**	**95.00**	31.19	47.38	55.50	24.90	43.35	54.49	36.71	53.62	61.75
Static Weight	45.00	66.00	76.50	61.88	78.75	87.50	25.88	43.19	55.31	**26.47**	**45.28**	**56.13**	39.81	58.31	68.86
Inst. Weight (Ours)	**60.00**	77.50	**87.50**	67.50	85.00	89.38	29.63	45.00	53.88	26.06	44.17	54.78	**45.80**	**62.92**	**71.39**
Gemma-4-E2B-it															
Final Layer	24.50	42.50	52.00	30.63	48.13	59.38	17.06	30.00	37.69	12.01	21.70	29.72	21.05	35.58	44.70
Mean Layer	12.50	25.00	37.00	52.50	69.38	78.13	**32.00**	**46.06**	**54.63**	11.63	21.65	29.38	27.16	40.52	49.79
Inst. Weight (Ours)	**25.50**	**44.50**	**54.50**	**58.75**	**74.38**	**79.38**	28.63	43.81	52.31	**15.07**	**26.94**	**36.08**	**31.99**	**47.41**	**55.57**

Table 17: Full Recall@K for audio layer aggregation on INSPIRE. Avg.: macro-average across the four subsets. Inst. Weight denotes instruction-aware audio layer weighting. Static Weight uses trainable weights independent of instructions and prompts. Bold marks column maxima within each backbone.

Layer Aggregation	DailyTalk	VCTK	Expresso				Synthetic			
	Cont.	Spk.	Spk.	Sty.	Spk. +Sty.	Spk. +Sty.*	Ans.	Spk.	Sty.	Env.
Qwen2.5-Omni-3B										
Layer 13	15.50	76.88	44.75	**45.50**	**67.25**	**38.25**	2.00	43.50	**54.75**	39.25
Final Layer	**82.50**	48.75	11.25	39.75	44.75	17.25	**94.50**	27.50	40.50	31.00
Mean Layer	70.00	71.25	37.75	42.00	57.75	35.00	87.75	50.75	45.50	40.50
Finetuned	32.50	**91.25**	**51.50**	42.25	65.00	30.75	68.25	52.50	54.50	43.25
Static Weight	66.00	78.75	33.00	41.75	61.50	36.50	87.00	48.50	52.25	**45.50**
Inst. Weight (Ours)	77.50	85.00	41.25	42.00	62.00	34.75	90.50	**53.50**	53.75	35.50
Gemma-4-E2B-it										
Final Layer	42.50	48.13	13.50	45.50	43.25	17.75	**83.25**	28.75	37.50	7.75
Mean Layer	25.00	69.38	32.75	42.25	**72.50**	**36.75**	1.50	**54.50**	44.50	4.00
Inst. Weight (Ours)	**44.50**	**74.38**	**33.00**	**48.00**	63.75	30.50	80.58	51.25	**47.25**	**8.50**

Layer Aggregation	Synthetic										
	Spk. +Sty.	Spk. +Sty.*	Spk. +Env.	Spk. +Env.*	Sty. +Env.	Sty. +Env.*	Sty.* +Env.	Spk. +Sty. +Env.	Spk. +Sty.* +Env.	Spk. +Sty. +Env.*	Spk. +Sty.* +Env.*
Qwen2.5-Omni-3B											
Layer 13	44.50	40.50	46.00	15.00	37.75	8.50	34.50	63.00	58.50	22.50	18.75
Final Layer	30.75	27.25	35.75	19.50	28.50	14.00	29.50	45.75	38.50	32.50	25.25
Mean Layer	49.75	43.75	**55.00**	20.75	35.75	16.50	38.75	65.50	56.25	**38.75**	**28.25**
Finetuned	49.75	43.50	51.25	19.50	35.75	11.00	**39.50**	**67.50**	**62.75**	32.50	18.75
Static Weight	48.25	41.25	50.25	**22.00**	**40.50**	**18.25**	37.25	67.00	57.25	37.25	26.75
Inst. Weight (Ours)	**54.25**	**46.00**	47.50	17.25	35.75	15.00	38.00	61.75	57.00	30.00	26.75
Gemma-4-E2B-it											
Final Layer	30.50	24.00	14.50	**9.25**	12.25	**5.00**	**10.50**	24.00	13.25	14.00	11.00
Mean Layer	**55.50**	**50.00**	13.25	6.00	4.50	3.75	3.50	29.25	**25.00**	14.00	**15.50**
Inst. Weight (Ours)	47.00	45.25	**15.50**	7.25	**13.50**	1.25	8.25	**31.50**	18.25	**15.00**	13.75

Table 18: Recall@50 by instruction type for audio layer aggregation on INSPIRE. Inst. Weight denotes instruction-aware audio layer weighting. Static Weight uses trainable weights independent of instructions and prompts. Instruction labels follow Table 13; bold marks column maxima within each backbone.

Audio Projection	DailyTalk			VCTK			Expresso			Synthetic			Avg.		
	R@10	R@50	R@100	R@10	R@50	R@100	R@10	R@50	R@100	R@10	R@50	R@100	R@10	R@50	R@100
Qwen2.5-Omni-3B															
Frozen	**69.00**	**83.00**	**87.50**	24.38	41.88	56.88	14.13	26.50	36.50	16.83	30.80	42.00	31.09	45.55	55.72
Trainable	59.50	82.50	85.50	**34.38**	**48.75**	**59.38**	**15.88**	**28.25**	**36.88**	**18.47**	**34.72**	**45.20**	**32.06**	**48.56**	**56.74**
Gemma-4-E2B-it															
Frozen	17.00	34.00	46.00	24.38	45.00	58.13	13.63	24.69	31.75	9.06	15.83	21.78	16.02	29.88	39.42
Trainable	**24.50**	**42.50**	**52.00**	**30.63**	**48.13**	**59.38**	**17.06**	**30.00**	**37.69**	**12.01**	**21.70**	**29.72**	**21.05**	**35.58**	**44.70**

Table 19: Recall@K on INSPIRE using only the final audio-encoder hidden states. Avg.: macro-average across the four subsets. Frozen and Trainable indicate whether the audio projection is frozen or updated during retrieval training. Bold marks column maxima within each backbone.

Audio Projection	DailyTalk	VCTK	Expresso				Synthetic			
	Cont.	Spk.	Spk.	Sty.	Spk. +Sty.	Spk. +Sty.*	Ans.	Spk.	Sty.	Env.
Qwen2.5-Omni-3B										
Frozen	**83.00**	41.88	**14.50**	34.25	39.25	**18.00**	92.25	24.00	39.75	29.25
Trainable	82.50	**48.75**	11.25	**39.75**	**44.75**	17.25	**94.50**	**27.50**	**40.50**	**31.00**
Gemma-4-E2B-it										
Frozen	34.00	45.00	9.75	35.25	37.75	16.00	**84.75**	17.25	**39.25**	0.75
Trainable	**42.50**	**48.13**	**13.50**	**45.50**	**43.25**	**17.75**	83.25	**28.75**	37.50	**7.75**

Audio Projection	Synthetic										
	Spk. +Sty.	Spk. +Sty.*	Spk. +Env.	Spk. +Env.*	Sty. +Env.	Sty. +Env.*	Sty.* +Env.	Spk. +Sty. +Env.	Spk. +Sty.* +Env.	Spk. +Sty. +Env.*	Spk. +Sty.* +Env.*
Qwen2.5-Omni-3B											
Frozen	26.75	23.00	30.00	17.00	26.50	12.00	28.50	39.00	30.25	25.75	18.00
Trainable	**30.75**	**27.25**	**35.75**	**19.50**	**28.50**	**14.00**	**29.50**	**45.75**	**38.50**	**32.50**	**25.25**
Gemma-4-E2B-it											
Frozen	26.25	19.50	5.00	3.50	3.00	2.50	3.75	11.25	6.25	6.50	8.00
Trainable	**30.50**	**24.00**	**14.50**	**9.25**	**12.25**	**5.00**	**10.50**	**24.00**	**13.25**	**14.00**	**11.00**

Table 20: Recall@50 by instruction type on INSPIRE using only the final audio-encoder hidden states. Frozen and Trainable indicate whether the audio projection is frozen or updated during retrieval training. Instruction labels follow Table 13. Bold marks column maxima within each backbone.